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Advanced Phishing Detection Using Machine Learning with Feature Selection, Risk Assessment, and Explainable AI

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Abstract

Abstract Phishing attacks continue to pose a major cybersecurity threat by tricking users into revealing sensitive information such as login credentials and financial data through fraudulent websites. This paper presents an advanced phishing URL detection framework using machine learning techniques to improve detection accuracy, efficiency, and interpretability. The proposed system employs permutation-based feature selection to reduce dimensionality and identify the most influential URL features. An Extreme Gradient Boosting (XGBoost) classifier is then used to analyze these selected features and classify URLs as phishing or legitimate. To enhance practical cybersecurity applications, a probability-based risk assessment module is integrated to categorize detected URLs into low, medium, and highrisk levels. Additionally, Explainable Artificial Intelligence (XAI) techniques using SHAP (SHapley Additive explanations) are incorporated to provide transparent insights into the model's decision-making process. The trained model and preprocessing pipeline are deployed in a web-based application for real-time phishing detection. Experimental results demonstrate that the proposed system achieves high classification accuracy while maintaining efficient prediction speed and model interpretability, making it suitable for realworld cybersecurity environments

Keywords— Phishing URL Detection, Machine Learning, Gradient Boosting, Feature Optimization, Explainable AI, SHAP Analysis, Risk Classification, Web Security, Cyber Threat Detection, URL Analysis.

1. Introduction:

Phishing attacks remain a significant cybersecurity concern, where malicious actors design deceptive websites to obtain sensitive information such as user credentials and financial data. Conventional detection approaches, including blacklist and rule-based techniques, often fail to identify newly emerging phishing websites and dynamic attack patterns.

To address these limitations, machine learning-based approaches can effectively analyze URL characteristics and detect patterns that differentiate malicious websites from legitimate ones. In this study, a phishing detection framework is developed using feature selection and an XGBoost classification model to enhance prediction accuracy. The system further integrates a probability-based risk assessment mechanism along with Explainable Artificial Intelligence (SHAP) to provide interpretable and transparent predictions.

The proposed model is implemented as a web-based application for real-time URL analysis. The experimental evaluation indicates that the system achieves high detection performance with efficient computation, making it suitable for real-world cybersecurity applications.

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2.Literature Survey:

Phishing detection has become an important research area in cybersecurity due to the rapid growth of online services and increasing cyber threats. Phishing attacks aim to deceive users by creating fake websites that resemble legitimate platforms, leading to the disclosure of sensitive information such as passwords and financial details. As these attacks continue to evolve, developing effective detection mechanisms has become essential.

Early approaches to phishing detection mainly relied on blacklist-based techniques and heuristic rules. Blacklist methods compare URLs with a database of known phishing websites, while heuristic approaches use predefined patterns to identify suspicious URLs. Although these methods are effective for detecting previously known threats, they are unable to identify newly generated phishing websites and zero-day attacks.

To overcome these limitations, researchers have explored machine learning techniques for phishing detection. Various algorithms such as Support Vector Machines (SVM), Decision Trees, and Random Forest have been applied to classify URLs based on extracted features. These models analyze characteristics such as URL length, domain structure, presence of special characters, and security indicators to differentiate between phishing and legitimate websites.

Several studies have shown that feature engineering plays a crucial role in improving phishing detection performance. Extracting relevant features from URLs and reducing redundant features can significantly enhance model accuracy and efficiency. Feature selection techniques help in identifying the most important attributes that contribute to classification, thereby reducing computational complexity.

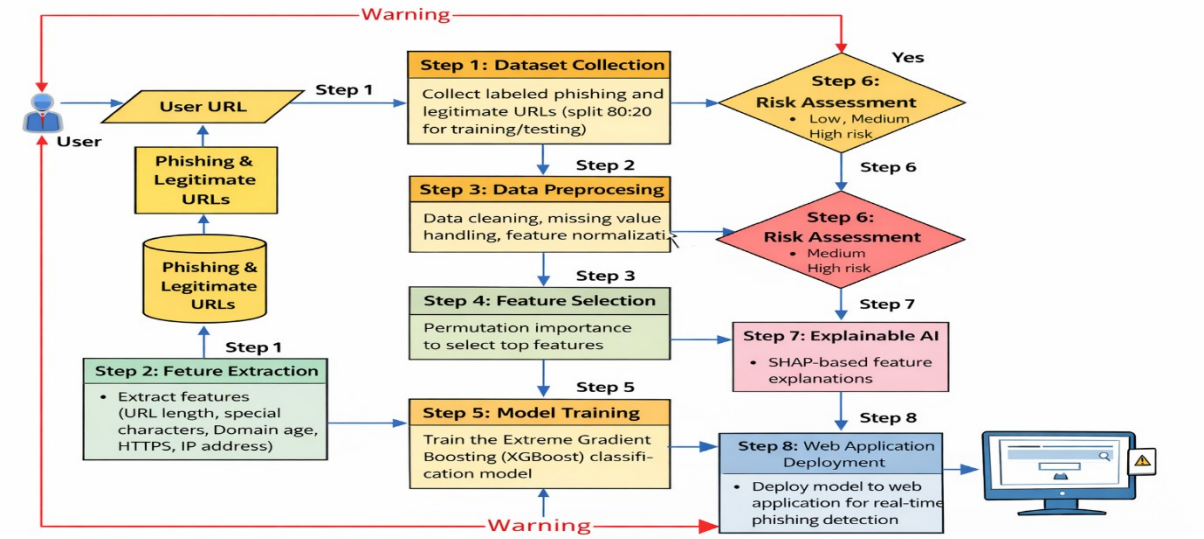
Recent research has focused on advanced ensemble learning methods such as Extreme Gradient Boosting (XGBoost), which combines multiple decision trees to improve prediction accuracy. These models have demonstrated superior performance compared to traditional machine learning approaches due to their ability to handle complex data patterns and provide better generalization.

In addition to accuracy, interpretability has become an important aspect of modern machine learning systems. Explainable Artificial Intelligence (XAI) techniques such as SHAP have been introduced to provide insights into model predictions. These methods help in understanding how different features influence the classification outcome, thereby improving transparency and trust in the system.

Despite these advancements, many existing studies primarily focus on improving classification accuracy and often overlook aspects such as feature optimization, risk assessment, and interpretability. Therefore, this research proposes an integrated phishing detection framework that combines feature selection, XGBoost classification, risk assessment, and explainable AI to achieve both high performance and model transparency.

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3.METHODOLOGY



3.1 Data Collection

The dataset used in this study consists of labeled phishing and legitimate URLs collected from publicly available cybersecurity sources. Each URL is associated with multiple features that describe its structural and domain characteristics. The dataset is prepared to ensure a balanced distribution of malicious and benign samples for effective model training and evaluation.

3.2 Feature Extraction

Relevant features are extracted from each URL to capture patterns associated with phishing attacks. These features include structural properties and security-related indicators.

Typical extracted features include:

- URL length
- Number of special characters
- Presence of HTTPS protocol
- Number of subdomains
- Domain age
- Presence of IP address in URL

These features are used as input variables for the machine learning model.

3.3 Data Preprocessing

Before training the model, the dataset undergoes preprocessing to improve data quality and consistency. Missing or inconsistent values are handled appropriately. All numerical features are normalized using the MinMaxScaler technique, which scales values within the range of 0 to 1.

This step ensures that all features contribute equally during model training and improves convergence of the learning algorithm.

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3.4 Feature Selection

Feature selection is performed to identify the most relevant features and reduce redundancy in the dataset. In this study, permutation importance with a Random Forest model is used to evaluate feature significance.

The top-ranked features are selected based on their importance scores. This reduces computational complexity and improves the efficiency and accuracy of the classification model.

3.5 Model Development

The phishing detection model is developed using the Extreme Gradient Boosting (XGBoost) algorithm. XGBoost is an ensemble learning technique that combines multiple decision trees to improve predictive performance.

The dataset is divided into training and testing sets using an 80:20 ratio. The model learns patterns from the training data and predicts whether a given URL is phishing or legitimate.

3.6 Risk Assessment Module

A probability-based risk assessment mechanism is incorporated into the system. The trained model outputs a probability score indicating the likelihood of a URL being phishing.

Based on this score, URLs are categorized into three risk levels:

- Low Risk – low probability of phishing
- Medium Risk – moderate probability
- High Risk – high probability of phishing

This classification helps users understand the severity of potential threats.

3.7 Explainable AI Integration

To enhance model transparency, Explainable Artificial Intelligence (XAI) techniques are applied using SHAP (SHapley Additive exPlanations).

SHAP provides feature-level explanations by showing how each feature contributes to the final prediction. This allows users to interpret the model's decisions and increases trust in the system.

3.8 System Deployment

The trained model is deployed as a web-based application for real-time phishing detection. Users can input URLs into the system, and the application returns the classification result along with the corresponding risk level and explanation.

4.Results and Discussion

4.1 Overall Performance

The proposed phishing detection framework was evaluated using multiple machine learning models to compare performance. The results indicate that the XGBoost classifier outperforms other models in terms of accuracy and detection capability.

Table 2 presents the comparative performance of different models.

Table 2. Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1 Score
SVM	0.92	0.91	0.90	0.90
Random Forest	0.94	0.93	0.94	0.93
XGBoost(Proposed)	0.966	0.95	0.96	0.95

The results highlight that XGBoost achieves the highest accuracy of **96.6%**, demonstrating superior capability in identifying phishing URLs.

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4.2 Model Performance Analysis

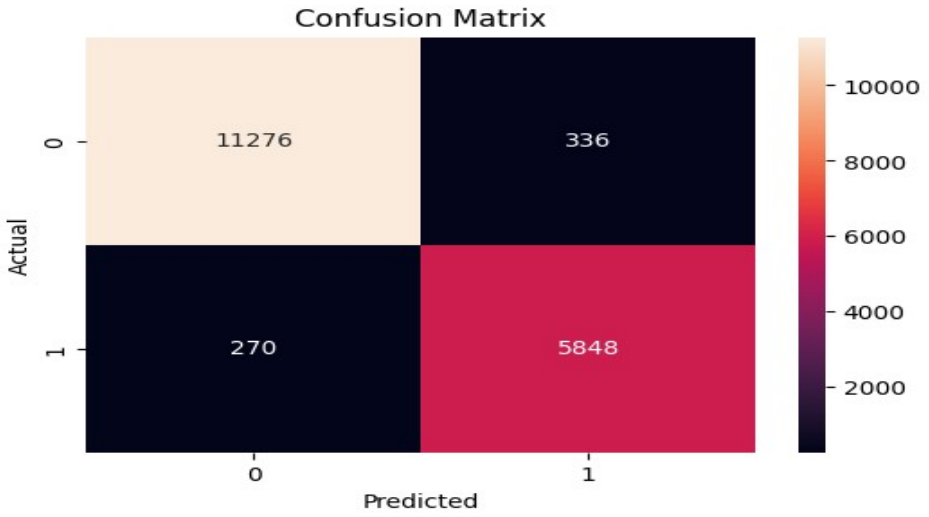
The performance metrics confirm the effectiveness of the proposed model. A precision of 0.95 indicates accurate identification of phishing URLs, while a recall of 0.96 shows strong detection capability. The F1 score of 0.95 reflects a balanced trade-off between precision and recall.These results indicate that the proposed model provides reliable and accurate phishing detection.

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.97	0.97	11612
1	0.95	0.96	0.95	6118
accuracy			0.97	17730
macro avg	0.96	0.96	0.96	17730
weighted avg	0.97	0.97	0.97	17730

4.3 Confusion Matrix Analysis

The confusion matrix further validates the model performance. Most URLs are correctly classified, with only a small number of misclassifications. The model shows low false positives and minimal false negatives, indicating strong detection reliability.



4.4 Discussion

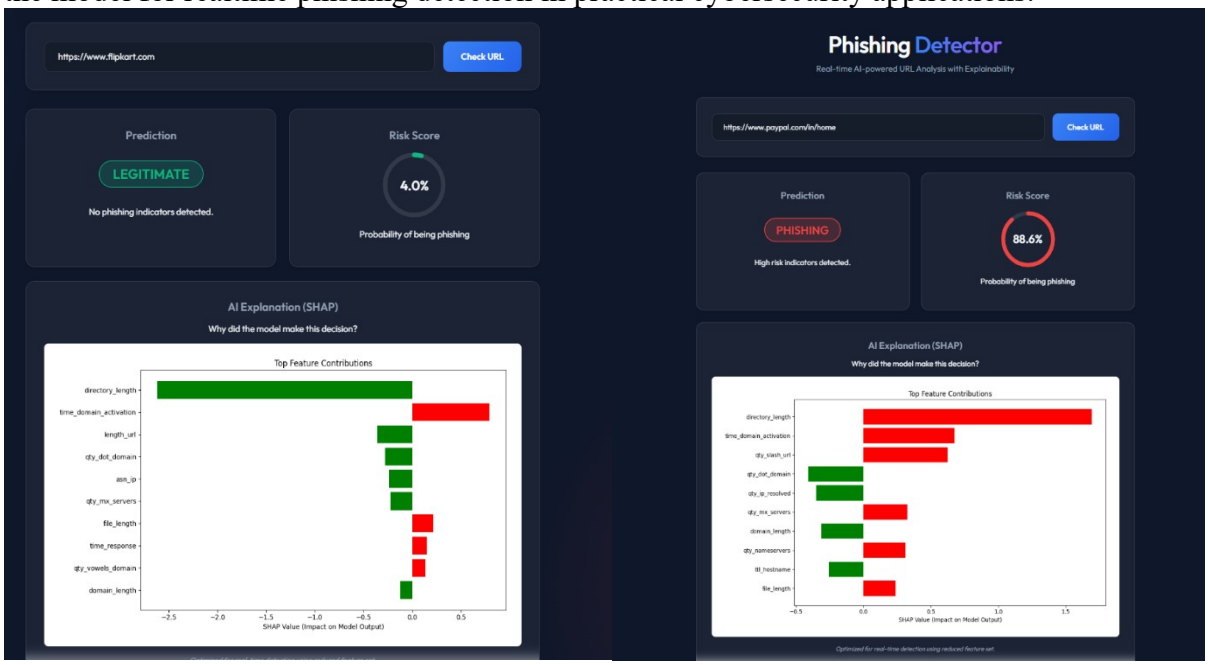
The results demonstrate that integrating feature selection with XGBoost improves detection accuracy while reducing computational complexity. Additionally, the use of SHAP-based explainability provides insights into feature contributions, enhancing transparency and user trust.

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Overall, the proposed approach offers an effective and practical solution for phishing detection in real-world cybersecurity applications.

5.Conclusion

This research proposed a machine learning–based framework for detecting phishing URLs using feature selection and an XGBoost classification model. The system analyzes structural URL features to distinguish phishing websites from legitimate ones and provides risk-level assessment for better threat understanding. Experimental results show that the proposed model achieves high performance with an accuracy of 96.6%, precision of 0.95, recall of 0.96, and F1 score of 0.95, demonstrating its effectiveness in detecting phishing attacks. Additionally, the integration of Explainable Artificial Intelligence (SHAP) improves transparency by explaining how different features influence the model's predictions. In future work, the system can be extended by incorporating webpage content analysis and deploying the model for realtime phishing detection in practical cybersecurity applications.



6.Future Work

Future work can extend the proposed phishing detection system in several directions. First, additional features such as webpage content, HTML structure, and real-time threat intelligence can be incorporated to improve detection accuracy. Second, deep learning models can be explored to capture more complex patterns in phishing URLs.

Furthermore, the system can be deployed as a browser extension or mobile application to provide real-time protection for users. Improving model efficiency for large-scale and real-time environments is another important area of future research. Finally, continuous model updating using new phishing data can enhance adaptability against evolving cyber threats.

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